MobiData BW MCP – Data Coverage

Snapshot: 2026-10-17, from the live CKAN catalogue (`https://mobidata-bw.de/api/3/action/`), live ParkAPI and live GeoServer WFS.

Status legend

Status	Meaning
Covered	A purpose-built tool consumes this dataset (or the feed that is this dataset).
Generic	Reachable only through the generic `query_layer` (WFS) or `search_datasets` / `get_dataset` (metadata). No tailored tool.
Indirect	The same real-world data arrives through a different or aggregated source. This exact dataset is not read.
Not covered	Not read by any tool. `search_datasets` / `get_dataset` still return its metadata and download URLs.

Why a dataset is not covered (reason codes)

The server answers location-based questions with live or fresh data and a source, licence and freshness stamp. A dataset is left out when it does not fit that purpose, or when it cannot be served reliably. Every non-covered row in section 3 carries one or more of these codes.

Code	Reason
R1	Statistics, survey or analysis product (annual counts, studies, yearbooks, demand maps). It does not answer "what is near me / what is happening now", and its freshness is months to years.
R2	Bulk file with no API (XLSX, PDF, shapefile, ZIP, CSV). Serving it needs a download, parsing and storage pipeline per dataset, with no live data.
R3	Redundant. A subset or bundle of a feed that is already served; adding it would risk inconsistent answers.
R4	Single city, district or pilot. Low reach for the integration and maintenance cost.
R5	Access needs registration. TRIAS requires a personal requestor key from NVBW (request by email to mobidata-bw@nvbw.de).
R6	Data too thin to answer questions. Verified 2026-10-17: `mobilstationen-bw` has 1,972 polygons with no attributes; `bruecken-bw` has 9,825 points with only name, road and station (no clearance or load limit).
R7	Broken upstream resource. Verified 2026-10-17: the download URL returns HTTP 404.
R8	Portal or web-app link only; no machine-readable resource to fetch.
R9	The same data is already served through another channel.
R10	Valid candidate, deliberately deferred. Small or niche value, can be added later as a focused tool.

1. Category to code map

All 10 website categories exist as CKAN groups. 8 of 10 have at least one dedicated tool. "Other" and "Traffic counting data" have none.

#	Category (CKAN group)	Datasets	MCP tools	Code files
1	Accessibility (`barrierefreiheit`)	4	`stop_accessibility`, `find_accessible_parking`	tools/accessibility.py, clients/bfrk.py, clients/parkapi.py
2	E-mobility (`e-mobilitaet`)	3	`find_charging`, `find_sharing`	tools/charging.py, tools/sharing.py
3	Foot and bike traffic (`fuss-und-radverkehr`)	11	`find_parking` (`purpose=BIKE`), `find_sharing`, `stop_accessibility`, `query_layer`	tools/parking.py, tools/sharing.py, tools/spatial.py
4	Parking space (`parkraum`)	15	`find_parking`, `find_accessible_parking`, `data_health`	tools/parking.py, tools/accessibility.py, tools/health.py, clients/parkapi.py
5	Sharing offers (`sharing-angebote`)	8	`find_sharing`, `data_health`	tools/sharing.py, tools/health.py
6	Other (`sonstiges`)	10	none (metadata only)	tools/discovery.py
7	Traffic events (`verkehrsereignisse`)	2	`get_roadworks`, `find_traffic_messages`, `check_route`	tools/roadworks.py, tools/traffic_messages.py, tools/route_check.py, clients/static_files.py, clients/traffic_messages.py, clients/routing.py
8	Transport networks (`verkehrsnetze`)	6	`find_traffic_signs`, `check_route`, `query_layer`	tools/signs.py, tools/route_check.py, tools/spatial.py
9	Traffic counting data (`verkehrszaehldaten`)	14	none (metadata only)	tools/discovery.py
10	Public transport (`oeffentlicher-personenverkehr`)	24	`find_transit`, `get_departures`, `stop_accessibility`, `query_layer`	tools/transit.py, clients/gtfs.py, clients/wfs.py

Cross-cutting files: tools/discovery.py (`search_datasets`, `get_dataset`, `list_layers` for all 146 catalogue datasets), tools/spatial.py (`query_layer` for any of the 13 WFS layers), clients/ckan.py, clients/geocode.py.

2. Upstream sources actually fetched

Source	Used by	Config
ParkAPI v3 (71 upstream sources, 32,025 records)	`find_parking`, `find_accessible_parking`, `data_health`	`parkapi_url`
GeoServer WFS `MobiData-BW:charge_points`	`find_charging`, `data_health`	`wfs_url`
GeoServer WFS `sharing_stations`, `sharing_vehicles`	`find_sharing`, `data_health`	`wfs_url`

Source	Used by	Config
GeoServer WFS `transit_stops`, `transit_stations_with_served_routes`	`find_transit`	`wfs_url`
GTFS arrivals/departures API	`get_departures` (scheduled only)	`gtfs_url`
`roadworks_geojson.json`	`get_roadworks`, `check_route`	`roadworks_url`
`incidents_lmsbw.datex2.xml` (DATEX II traffic messages, about 35 situations, refreshed every few minutes)	`find_traffic_messages`, `check_route`	`traffic_messages_url`
`VZ-Daten/signs.geojson`	`find_traffic_signs`, `check_route`	`signs_url`
BFRK CSV files (NVBW, resolved through CKAN)	`stop_accessibility`, `find_accessible_parking`	`ckan_url`
OpenRouteService HGV directions	`check_route`	clients/routing.py
Nominatim	location resolution	`nominatim_url`
CKAN `package_search` / `package_show`	`search_datasets`, `get_dataset`, BFRK resource lookup	`ckan_url`

The remaining 8 of 13 WFS layers (`roadworks`, `park-api_car`, `park-api_car_lines`, `park-api_car_polygons`, `park-api_bicycle`, `radvis_cycle_network`, `bicycle_service_points`, `transit_shapes_with_routes`) are only reachable through `query_layer`. The first five duplicate data already served by dedicated tools. Spot-tested on 2026-10-17: `radvis_cycle_network`, `bicycle_service_points`, `transit_shapes_with_routes`, `park-api_bicycle` all return features.

3. Dataset-level coverage

3.1 Accessibility (4)

Dataset	Status	How / why
`bfrk-barrierefreiheit-an-bw-bahnhoefen` (SPNV stations)	Covered	`stop_accessibility`, `find_accessible_parking` read the BFRK CSVs
`bfrk-barrierefreiheit-an-bw-haltestellen` (ÖPNV stops)	Covered	`stop_accessibility`
`daten-barrierefreiheit` (BFRK API)	Indirect (R9)	Same BFRK survey data is read from the CSV exports instead of the API, which avoids a second integration for identical content
`fgue-bw` (pedestrian crossings)	Not covered (R10)	Small GeoJSON of crossing points; useful for step-free routing but too thin for a standalone tool

3.2 E-mobility (3)

Dataset	Status	How / why
`e-ladesaulen`	Covered	`find_charging` (WFS `charge_points`)
`e-kleinkraftrad-sharing`	Covered	`find_sharing` (`kind=moped`)
`fahrrad-e-ladesaulen` (Reutlingen)	Not covered (R7, R4)	Download URL returns HTTP 404, and it is a single-city dataset

3.3 Foot and bike traffic (11)

Dataset	Status	How / why
`gebundelte-fahrradabstellanlagen-baden-wurttemberg`	Covered	`find_parking` with `purpose=BIKE` (ParkAPI)
`bikesh`	Covered	`find_sharing` (`kind=bicycle`)
`cargobike`	Covered	`find_sharing` (`kind=cargo_bicycle`)
`lastenrad-sharing-lastenkarle`	Covered	Feed `lastenkarle` present in sharing layers (18 stations)
`bfrk-barrierefreiheit-an-bw-bahnhoefen`	Covered	`stop_accessibility`
`radnetz-bw`	Generic	Cycle-network geometry via `query_layer` on `radvis_cycle_network`. A dedicated tool would need routing logic, not a lookup
`radservicepunkte`	Generic	Repair/pump points via `query_layer` on `bicycle_service_points`; small point set that does not justify its own tool
`eco-counter-fahrradzahler`	Not covered (R1, R10)	Bike counter time series (CSV/API). A statistics question, not a "where" question; candidate for a later tool
`besserradfahren-swr-umfrage`	Not covered (R1, R2)	One-off SWR survey as XLSX
`fgue-bw`	Not covered (R10)	See 3.1
`fahrradparken-kapazitaet` (VRN)	Not covered (R4)	Regional ArcGIS layer for the VRN area only; no VRN bike source exists in ParkAPI

3.4 Parking space (15)

Dataset	Status	How / why
`gebuendelte-parkdaten-bw`	Covered	`find_parking` (ParkAPI, car)
`gebundelte-daten-strassen-parkplatze-bw`	Covered	`find_parking`; 22,169 on-street records, all static (not live)

Dataset	Status	How / why
`gebundelte-fahrradabstellanlagen-baden-wurttemberg`	Covered	`find_parking` with `purpose=BIKE`
`bfrk-barrierefreiheit-an-bw-bahnhoefen`	Covered	BFRK `Parkplatz.csv` via `find_accessible_parking`
`bfrk-barrierefreiheit-an-bw-haltestellen`	Covered	Stop accessibility only; its parking is not read
`p-m-parkplatze-baden-wurttemberg`	Covered	ParkAPI sources `pum_bw`, `p_m_bw`
`parkdaten-freiburg`	Covered	ParkAPI sources `freiburg`, `freiburg_scanner`
`parkdaten-herrenberg`	Covered	ParkAPI sources `herrenberg`, `herrenberg_bike`
`parkleitsystem-konstanz`	Covered	ParkAPI source `konstanz` (+ `konstanz_bike`)
`on-street-parkplatze` (Heidelberg)	Indirect (R9)	Heidelberg on-street comes from ParkAPI source `heidelberg_easypark` (7,969 records), so the city is answered, but not from this city dataset
`motorradparkplatze-pforzheim`	Not covered (R7)	Download URL returns HTTP 404. ParkAPI `pforzheim` has car parks only; the only motorcycle source is `esslingen` (43 records)
`ruhender-verkehr-s-bad-cannstatt`	Not covered (R4)	Static GeoJSON for one Stuttgart district; no ParkAPI source
`parkleitsystem-der-stadt-basel`	Not covered (R4)	Swiss city outside Baden-Württemberg; the upstream endpoint returned HTTP 500 on test
`fahrradparken-kapazitaet` (VRN)	Not covered (R4)	See 3.3
`verkehr` (Konstanz ESRI hub)	Not covered (R8)	Web-app and portal link only; the machine-readable Konstanz parking data is already served through ParkAPI

3.5 Sharing offers (8)

All 8 are bundled GBFS feeds that reach us through the WFS sharing layers, so `find_sharing` covers them.

Dataset	Status	How / why
`carsharing`	Covered	`find_sharing` (`kind=car`)
`bikesh`	Covered	`find_sharing` (`kind=bicycle`)
`cargobike`	Covered	`find_sharing` (`kind=cargo_bicycle`)
`escootch`	Covered	`find_sharing` (`kind=scooter`)
`e-kleinkraftrad-sharing`	Covered	`find_sharing` (`kind=moped`)
`stadtmobil-stuttgart-stationsdaten`	Covered	Feed `stadtmobil_stuttgart` (353 stations)

Dataset	Status	How / why
`swu2go-gbfs`	Covered	Feed `swu2go` (29 stations)
`lastenrad-sharing-lastenkarle`	Covered	Feed `lastenkarle` (18 stations)

3.6 Other (10)

No category-specific tool. All 10 are Not covered; metadata and links remain available through `search_datasets` and `get_dataset`.

Dataset	Status	Why
`posterumfrage-angebote-mobilitatsstation-eselsberg-2107`	Not covered (R1, R4)	One-off survey in Ulm (CSV)
`ferdinand-steinbeis-institut-pilotstudie-track4science`	Not covered (R1)	Research study data (CSV)
`motorradlarm-hotspots-daten`	Not covered (R1, R2)	Noise hotspot statistics as XLSX
`winterdienstsensor-neulingen-olbronn-durrn`	Not covered (R4)	Two-site sensor pilot; the resource link is not a directly fetchable HTTP URL
`statistisches-jahrbuch-verkehr`	Not covered (R1, R2)	Heidelberg statistics yearbook chapter (PDF/XLSX)
`motorradparkplatze-pforzheim`	Not covered (R7)	Download URL returns HTTP 404 (see 3.4)
`mobilstationen-bw`	Not covered (R6)	1,972 polygons with no attributes: no name, no services, no operator. It could only answer "there is a hub polygon here"; sharing stations already give the actual services
`gps-daten-wochenmarkt-lieferdienst`	Not covered (R1, R4)	GPS tracks of one delivery pilot in Ulm
`gebundelte-taxistellplatze-baden-wurttemberg`	Not covered (R7)	Download URL returns HTTP 404
`bruecken-bw`	Not covered (R6)	9,825 bridge points with name, road and station only. There is no clearance or load limit, so it cannot support truck route checks

3.7 Traffic events (2)

Dataset	Status	How / why
`baustelleninformationen-baden-wurttemberg`	Covered	`get_roadworks`, `check_route` (GeoJSON)

Dataset	Status	How / why
`meldung` (traffic messages, DATEX II)	Covered	`find_traffic_messages` and `check_route`. Accidents, congestion, closures and obstructions from the DATEX II feed; records with the same text and place are merged

3.8 Transport networks (6)

Dataset	Status	How / why
`verkehrszeichen-baden-wurttemberg`	Covered	`find_traffic_signs`, `check_route` (advisory)
`radnetz-bw`	Generic	See 3.3
`strassennetz-netzknoten-baden-wuerttemberg`	Not covered (R2)	Reference network and junction tables (CSV/XLS/ZIP); no live data and no question it answers directly
`bedarfsumleitungen-baden-wurttemberg`	Not covered (R2, R10)	Planned diversion routes as XLSX. Relevant to closures, but it needs the network geometry above to be usable
`vz-freiburg`	Not covered (R4)	Freiburg-only signs WFS; statewide signs are already covered
`fgue-bw`	Not covered (R10)	See 3.1

3.9 Traffic counting data (14)

No tool. All 14 are Not covered. These are traffic and passenger statistics: historic, periodic and mostly report files, so they do not fit a live, place-based assistant.

Dataset	Status	Why
`eco-counter-fahrradzahler`	Not covered (R1, R10)	Bike counter time series; the only one here with an API, so the best later candidate
`stundenwerte_dauerzaehlstellen`	Not covered (R1, R2)	Hourly values of permanent counters as bulk CSV (ZIP)
`ergebnisse_ganglinien_dauerzaehlstellen`	Not covered (R1, R2)	Result reports and curves (PDF/XLSX ZIP)
`endergebnisse_strassenverkehrszaehlung`	Not covered (R1, R2)	Final census results (PDF/XLSX ZIP)
`bundesweite_strassenverkehrszaehlung`	Not covered (R1, R2)	National census reports (PDF)
`verkehrsentwicklungen_jahresvergleiche_strassenverkehr`	Not covered (R1, R2)	Year-on-year comparison report (PDF ZIP)
`karte_strassenverkehrszaehlung`	Not covered (R1)	Census map visualisation and CSV
`nachfragekarte-spnv-bw`	Not covered (R1)	Rail demand analysis map (GeoJSON)
`automatische-fahrgast-zahldaten-baden-wurttemberg`	Not covered (R1, R2)	Passenger counts on regional rail (CSV ZIP)
`zaehldaten-gemeinde-berg`	Not covered (R1, R2, R4)	One municipality, XLSX
`zaehldaten-stadt-weingarten`	Not covered (R1, R2, R4)	One municipality, XLSX
`zaehldaten-gemeinde-baienfurt`	Not covered (R1, R2, R4)	One municipality, XLSX

Dataset	Status	Why
`zaehldaten-gemeinde-baindt`	Not covered (R1, R2, R4)	One municipality, XLSX
`zaehldaten-ravensburg`	Not covered (R1, R2, R4)	One municipality, XLSX

3.10 Public transport (24)

Dataset	Status	How / why
`soll-fahrplandaten-baden-wurttemberg` (no line shapes)	Covered	`find_transit` (WFS stops/stations), `get_departures` (GTFS API)
`bfrk-barrierefreiheit-an-bw-bahnhoefen`	Covered	`stop_accessibility`
`bfrk-barrierefreiheit-an-bw-haltestellen`	Covered	`stop_accessibility`
`soll-fahrplandaten-baden-wurttemberg-mit-linienverlauf`	Generic	Line shapes via `query_layer` on `transit_shapes_with_routes`; shapes are map geometry, not an answer to a rider question
`haltestellen-baden-wuerttemberg`	Indirect (R9)	Equivalent stop data comes from the GTFS-derived WFS layers
`soll-fahrplandaten-vvs`	Indirect (R3)	Overlaps the statewide GTFS; per-operator inclusion not verified
`soll-fahrplandaten-verkehrsverbunde-bw-einzeln-ohne-linienverlauf`	Indirect (R3)	Per-network subset of the statewide GTFS
`soll-fahrplandaten-verkehrsverbunde-bw-einzeln-mit-linienverlauf`	Indirect (R3)	Per-network subset of the statewide GTFS
`soll-fahrplandaten-s-und-u-bahnen-bw-ohne-linienverlauf`	Indirect (R3)	Rail subset of the statewide GTFS
`soll-fahrplandaten-s-und-u-bahnen-bw-mit-linienverlauf`	Indirect (R3)	Rail subset of the statewide GTFS
`opnv-fahrplandaten-der-swu`	Indirect (R3)	Ulm operator feed; overlaps the statewide GTFS, not verified
`gebundelte-soll-fahrplandaten-fur-baden-wurttemberg-rheinland-pfalz-und-saarland-ohne-linienverlauf`	Indirect (R3)	Wider bundle; only the BW part is needed and served
`gebundelte-fahrplandaten-fur-baden-wurttemberg-rheinland-pfalz-und-saarland-mit-linienverlauf`	Indirect (R3)	Wider bundle; only the BW part is needed and served
`daten-barrierefreiheit`	Indirect (R9)	BFRK CSV used instead of the API
`trias` (real-time timetable)	Not covered (R5)	Real-time departures would be the most valuable addition, but TRIAS needs a personal requestor key. Access is granted on request by email to mobidata-bw@nvbw.de. Until then `get_departures` is scheduled data only

Dataset	Status	How / why
`mitfahren` (Amarillo, GTFS-RT)	Not covered (R10)	Ride-matching offers rather than a stop or timetable lookup; a candidate for a later tool
`opnv-guteklassen`	Not covered (R1, R2)	Service-quality isochrones as shapefile; an analysis product
`nachfragekarte-spnv-bw`	Not covered (R1)	Rail demand analysis map (GeoJSON)
`zugkm-pkm-bw`	Not covered (R1, R2)	Annual train and passenger-kilometre statistics (XLSX)
`automatische-fahrgast-zahldaten-baden-wurttemberg`	Not covered (R1, R2)	Passenger count statistics (CSV ZIP)
`stops-20220624-geoexport` (VRN stops)	Not covered (R2, R4)	VRN-area stops as shapefile, dated 2022; statewide stops are already served
`linienverlauf-vrn`	Not covered (R2, R4)	VRN-area line shapes as shapefile
`gebundelte-taxistellplatze-baden-wurttemberg`	Not covered (R7)	Download URL returns HTTP 404
`verkehr` (Konstanz)	Not covered (R8)	Portal and web-app link only

4. Summary for the presentation

Counts are group memberships (97). A dataset in two groups is counted twice. 76 unique datasets sit in a group; the whole catalogue has 146.

Category	Total	Covered	Generic	Indirect	Not covered
Accessibility	4	2	0	1	1
E-mobility	3	2	0	0	1
Foot and bike traffic	11	5	2	0	4
Parking space	15	9	0	1	5
Sharing offers	8	8	0	0	0
Other	10	0	0	0	10
Traffic events	2	2	0	0	0
Transport networks	6	1	1	0	4

Category	Total	Covered	Generic	Indirect	Not covered
Traffic counting data	14	0	0	0	14
Public transport	24	3	1	10	10
Total	**97**	**32**	**4**	**12**	**49**

Reason frequency (counted from the rows above; a row can carry several codes):

Code	Rows	Applies to
R1 statistics, surveys, analysis	25	Not covered
R2 bulk files, no API	21	Not covered
R4 single city or pilot	16	Not covered
R10 deferred candidates	7	Not covered
R7 broken upstream URL (404)	5	Not covered
R6 data too thin	2	Not covered
R8 portal link only	2	Not covered
R5 registration required (TRIAS)	1	Not covered
R3 redundant bundle	8	Indirect
R9 same data via another channel	4	Indirect

Strongest coverage is for live, location-based questions (parking, sharing, charging, roadworks, signs, transit, accessibility). Statistical, survey, counting and network-planning datasets are not covered.

5. Caveats to state openly

- One ParkAPI endpoint is not one dataset. It aggregates 71 upstream sources, so a parking "dataset" is covered through its source, not as a file.
- On-street parking is returned but static. None of the 22,169 on-street records has live occupancy.
- `find_parking` fetches every site in the radius, sorts by distance and only then applies the limit, so the nearest sites are never cut off. In very dense areas (more than about 1,000 sites) it narrows the radius and says so in a warning.
- `get_departures` is scheduled data. No real-time public transport until a TRIAS requestor key is obtained (R5).
- Traffic messages are a state-wide DATEX II snapshot with German text. Records with the same text and place are merged. If the feed is down, `check_route` still returns roadworks and signs and says that incidents are missing.
- Motorcycle parking data exists for very few places, and `find_parking` says so when `purpose=MOTORCYCLE`.
- "Indirect" rows for GTFS bundles are inferred from overlap with the statewide feed, not verified operator by operator.